

Mohamed Elsherbini

Location: Doha, Qatar

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GPA: 3.89 / 4.00

Professional Summary

- High-performing **Mechanical Engineering** student focused on **Aerodynamics, Heat Engines, and Aerospace Applications**, with a **Computer Science** minor focused on **machine learning and database systems** and a **3.91/4.00 GPA**. Recognized on either the **Dean's List or Vice President's List** every semester since beginning university, consistently placing among the top students in the cohort.
- Experienced in **renewable-energy asset management, wind-turbine diagnostics, SCADA analysis, CFD, mechanical design, and failure assessment**. At **Iberdrola**, introduced a new turbine-analysis methodology with promising potential for detecting intermediate and catastrophic failures.
- Leader of a **Q-Chem-sponsored multidisciplinary senior design project**, coordinating more than **10 students** across Mechanical, Chemical, Electrical, and Computer Science disciplines to reduce VOC emissions from floating-roof storage tanks through CFD-based wind-shield design, vapor recovery, instrumentation, and digital verification.
- Award-winning innovator and researcher with experience in solar-panel fault monitoring, sustainable product development, and AI-driven parametric CAD, with particular interest in **sustainable energy, advanced engines, turbomachinery, aerospace engineering, and data-driven engineering**.

Education

Bachelor of Science in Mechanical Engineering Minor in Computer Science *Qatar University*

College of Engineering

Expected Graduation: 2027

GPA: 3.89 / 4.00

Mechanical Engineering focus: Aerodynamics, Heat Engines, and Aerospace Applications

Computer Science focus: Machine Learning and Database Systems

Engineering Experience & Research

Iberdrola

June 2026 – August 2026

Asset Management Engineering Intern

Location

- Developed a new data-driven methodology for detecting and assessing **intermediate and catastrophic wind-turbine blade failures**, demonstrating promising potential beyond the existing analysis methods used by the team.
- Analyzed large **SCADA and operational datasets** using **Python, Pandas, SQL, and Microsoft Excel** to evaluate turbine performance, abnormal trends, stoppages, downtime patterns, asset availability, and operational condition.
- Investigated wind-turbine failure modes and mechanisms to support **asset health assessment, predictive maintenance, and maintenance decision-making**.

Qatar University – Q-Chem Sponsored Senior Design Project

September 2026 – May 2027

Multidisciplinary Project Leader & Mechanical Design Engineer

Doha, Qatar

- Leading the early-stage development of an economically justified retrofit package for **Q-Chem external floating-roof tanks** to reduce summer **volatile organic compound emissions**.

- Coordinating more than **10 students** across Mechanical, Chemical, Electrical, and Computer Science disciplines and integrating mechanical design, hazardous-area instrumentation, SCADA monitoring, digital-twin verification, prototyping, safety, and economic assessment.
- Responsible for **CFD simulation of airflow around the tank** and the design and optimization of a wind-shield system to reduce effective wind velocity and wind-driven evaporative losses.
- Collaborating with Chemical Engineering students to design and develop a **vapor recovery unit (VRU)** for capturing and recovering hydrocarbon vapors.
- **Supervised by: Dr. Sadok Sassi and Dr. Ahmed Sleiti**

KEO International Consultants

June 2025 – August 2025

MEP Design Intern

Doha, Qatar

- Assisted in mechanical drafting, modeling, and design of HVAC, plumbing, and fire-protection systems using **Revit MEP** and **HAP**.
- Applied **IPC, ASHRAE, and NFPA** standards during system layout, coordination, and documentation.
- Supported multidisciplinary coordination and interpretation of construction drawings.

Qatar University

September 2025 – Present

Engineering Tutor (Part-Time)

Doha, Qatar

- Tutor students in Thermodynamics, Newtonian Mechanics, Mechanics of Materials, and Probability & Statistics.
- Deliver concept-driven explanations and exam-focused problem-solving support.

Qatar University

Ongoing

Innovation Grant Researcher – AI for Parametric CAD

Doha, Qatar

- Developing an **AI-driven system** for automated parametric CAD generation by integrating SolidWorks modeling, macro-based coding, Python pipelines, and structured geometry datasets.
- **Supervised by: Dr. Abdeldjalil Bennecer**

Qatar University

May 2024 – August 2024

Research Methodology Intern

Doha, Qatar

- Conducted experimental research in the Driving Simulation Lab and analyzed behavioral data using **Python**.
- Contributed to technical reports and academic poster presentations.

Technical Skills

Programming & Data	Python, Pandas, SQL, MATLAB, Java, Microsoft Excel
Engineering Software	SolidWorks, AutoCAD, Revit MEP, HAP, Aspen HYSYS, ANSYS
Engineering Methods	CFD, FEA, SCADA analysis, predictive maintenance, failure-mode analysis, numerical modeling, CAD automation

Awards & Innovation

QRDI x Iberdrola Innovation Competition

2025

2nd Place Winner – Solar Panel Micro-Crack Monitoring

Doha, Qatar

- Developed a cost-effective smart detection framework to identify the initiation and progression of photovoltaic micro-cracks and presented its technical feasibility and implementation strategy to industry evaluators.

Injaz Qatar

2025

Best Sustainable Idea Award – Shamsiya Project

Doha, Qatar

- Designed a fully solar-powered smart umbrella, validated its system concept using **Aspen HYSYS**, and created the mechanical model in **SolidWorks**.

Selected Projects

Mechanical Vibrations	Designed, manufactured, and tested a tuned vibration absorber for a 1 mm-thick cantilever beam, reducing vibration intensity by approximately 60% .
Machine Design	Designed and manufactured a mechanical machine capable of maintaining continuous revolutions for more than 30 seconds .
Mechanical Mechanisms	Designed and analyzed a hand-cranked multi-link lifting mechanism using kinematic, dynamic, gear-design, and prototype-validation methods.

Certifications & Honors

Academic Excellence	Dean's List or Vice President's List in every completed semester at Qatar University
BIM Training	BIM Modeler Training Program – Arabian Infotech
Revit Professional	Revit for Mechanical & Plumbing Design – LinkedIn Learning
Fire Protection	Revit: Sprinkler Design – LinkedIn Learning
Research	Research Methodology Internship Certificate – Qatar University

Languages

Arabic	Native
English	Fluent